In [1]:
```
import pandas as pd

df = pd.read_csv("C:/Users/mestr/Downloads/archive (10)/Churn_Modelling.csv")

print(df.head())
print(df.shape)
```

```
   RowNumber  CustomerId   Surname  CreditScore Geography  Gender  Age  \
0          1    15634602  Hargrave          619    France  Female   42
1          2    15647311      Hill          608     Spain  Female   41
2          3    15619304      Onio          502    France  Female   42
3          4    15701354      Boni          699    France  Female   39
4          5    15737888  Mitchell          850     Spain  Female   43

   Tenure    Balance  NumOfProducts  HasCrCard  IsActiveMember  \
0       2       0.00              1          1               1
1       1   83807.86              1          0               1
2       8  159660.80              3          1               0
3       1       0.00              2          0               0
4       2  125510.82              1          1               1

   EstimatedSalary  Exited
0        101348.88       1
1        112542.58       0
2        113931.57       1
3         93826.63       0
4         79084.10       0
(10000, 14)
```

In [2]:
```
print(df.columns)
print(df.isnull().sum())
```

```
Index(['RowNumber', 'CustomerId', 'Surname', 'CreditScore', 'Geography',
       'Gender', 'Age', 'Tenure', 'Balance', 'NumOfProducts', 'HasCrCard',
       'IsActiveMember', 'EstimatedSalary', 'Exited'],
      dtype='object')
RowNumber          0
CustomerId         0
Surname            0
CreditScore        0
Geography          0
Gender             0
Age                0
Tenure             0
Balance            0
NumOfProducts      0
HasCrCard          0
IsActiveMember     0
EstimatedSalary    0
Exited             0
dtype: int64
```

In [3]:
```
df = df.drop(['RowNumber', 'CustomerId', 'Surname'], axis=1)
```

In [4]:
```
df = pd.get_dummies(df, columns=['Geography', 'Gender'], drop_first=True, dtype=
```

In [5]:
```
print(df.head())
print(df.shape)
```

```
   CreditScore  Age  Tenure    Balance  NumOfProducts  HasCrCard  \
0          619   42       2       0.00              1          1
1          608   41       1   83807.86              1          0
2          502   42       8  159660.80              3          1
3          699   39       1       0.00              2          0
4          850   43       2  125510.82              1          1

   IsActiveMember  EstimatedSalary  Exited  Geography_Germany  \
0               1        101348.88       1                  0
1               1        112542.58       0                  0
2               0        113931.57       1                  0
3               0         93826.63       0                  0
4               1         79084.10       0                  0

   Geography_Spain  Gender_Male
0                0            0
1                1            0
2                0            0
3                0            0
4                1            0
(10000, 12)
```

In [6]:
```
X = df.drop('Exited', axis=1)
y = df['Exited']
```

In [7]:
```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [8]:
```
print(X_train.shape)
print(X_test.shape)
```

```
(8000, 11)
(2000, 11)
```

In [9]:
```
from sklearn.preprocessing import StandardScaler

sc = StandardScaler()

X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

In [10]:
```
from sklearn.neural_network import MLPClassifier

model = MLPClassifier(hidden_layer_sizes=(10,), max_iter=500, random_state=42)

model.fit(X_train, y_train)
```

Out[10]:

▾ MLPClassifier

▶ Parameters

In [11]:
```
y_pred = model.predict(X_test)
```

In [12]:
```
from sklearn.metrics import accuracy_score
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.866
```

In [13]:
```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.866
```

In [14]:
```
from sklearn.metrics import confusion_matrix

print(confusion_matrix(y_test, y_pred))
```

```
[[1535   72]
 [ 196  197]]
```

In [ ]:
```

```